



Green Impact Product Report

Product: Carbon Cube Chair

Version: v1.0

Batch Estimate: 1000 units

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Date: August 26th 2025

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Regen Report Impact Template v2.0



About this Report

This report is created from a document template was developed by the [Global Ecobrick Alliance](#) (GEA) for the listing of [regenerative](#) products on the [GoBrik Shop](#). All products that are added to the GoBrik Regen Store, must be accompanied by a Green Impact Report and must demonstrate a subtractive CO₂ and Plastic impacts. It is under this condition that products in the store are considered “Regenerative”. The Green Impact Product Template and a further explanation of the terms and concepts herein can be found a www.ecobricks.org/principles.

Green Impact Reports are generated by the product, and not the GEA GEA does not endorse, nor corroborate any of the information herein.

The Global Ecobrick Alliance

The GEA is a not-for-profit Earth Enterprise, that operates on regenerative principles. The GEA maintains the GoBrik platform and the Brikcoin manual blockchain. The GEA also maintains the GoBrik store as a space for regenerative products.







Table of Contents

About this Report.....	2
The Global Ecobrick Alliance.....	2
CO ₂ Impact.....	6
Processes that Generate CO ₂	6
Production.....	6
Materials.....	7
Server Usage.....	7
Shipping to Customers.....	7
Product Process that Sequester CO ₂	9
Bamboo Plantation.....	9
Ecobrick Plastic Sequestration CO ₂ Credit.....	9
Replacing a similar Industrial Product.....	9
CO ₂ Impacts.....	11
Plastic Impact.....	12
Plastic Production.....	12
Manufacturing.....	12
Total plastic impact: 13.6 kg plastic per 1000 the carbon cube chair.....	12
Marketing / Labeling.....	12
Shipping.....	12
Plastic Offsetting.....	13
Ecobrick Plastic Sequestration.....	13
Replacing Plastic Equivalent Products.....	13
Plastic Impacts.....	14

Towards Greening, Circular & Regenerative Design





CO₂ Impact

One the clearest ways to measure the regenerative impact of a product is by measuring how much carbon dioxide (CO₂) is produced by the various processes that make up its life-cycle. The impacts of the various processes behind the production, marketing, sale, consumption and disposal of a product all have measurable determinations of CO₂ equivalency (CO₂e) from the amount of oil, gas and electricity that they involve.

When fossil fuels are burned to power cars, trucks and airplanes or used to power the factories that produce our products, CO₂ is produced. CO₂ enters the atmosphere and oceans, contributing to the atmospheric green house effect and ocean acidification. Around the world and there are well established and a widely accepted means for evaluating the 'carbon footprint' of various activities by metric unit (i.e. transportation by airplane per kilometer). The calculations in this report are based on carbon coefficients that are documented in the footnotes of the report.

Processes that Generate CO₂

Production

- The cutting, drilling and routing of the carbon cube chair requires electricity. We estimate from our workshop electricity bill that 300 kwh of electricity are used to produce 1000 units. Based on stats from East Java where our production is based we estimate a [0.256 kg of CO₂ is produced per kWh of electricity](#).¹ This results in an impact of **79.5Kg** CO₂.
- The shipping of completed the carbon cube chair from East Java to Jogja by truck (219km) have an impact of 0.069Kg/km according to the [ECTA](#).² This works out to an impact of **15.33kg** for shipping 1000 units to Jogja.
- Production total: **38.73 Kg** CO₂e per batch of 1000 units

1 <https://bulb.co.uk/carbon-tracker/>

2 https://www.ecta.com/resources/Documents/Best Practices Guidelines/guideline_for_measuring_and_managing_CO2.pdf



Materials

- Glue and finishing oils we use have CO₂ production impacts. [Crownpaints estimates the CO₂ impact of a 5L can to be 13.58 kg CO₂e.](#)³ We estimate that we need one 250ml can per 25, or 40 can per 1000 units. This 10L or epoxy paint, for 1000 Carbon Cube Chairs.
- Materials Total: **27.16** kg of CO₂e per batch of 1000 units

Server Usage

The purchase of a Carbon Cube chair on the GEA Regen Store server involves the visiting and use of various server resources. We measure the impact of these using the [websitecarbon.com](#) tool.^{4 5}

- The the ecojoiner profile page: 0.10 g of CO₂
- The the carbon cube chair Regen Report (300kb): 0.05 g of CO₂
- Regen Store product page loading: 0.10 g of CO₂ (estimated)
- Buwana login process: 0.05 g of CO₂
- Purchase process on regen store: 0.10 g of CO₂
- **Total:** 0.5g of CO₂ per purchase
- Server Total per 1000 units sold: **0.5Kg** of CO₂e

Shipping to Customers

- We're beginning with only Indonesian sales, and only shipping packages by domestic JNE shipping. The net weight of the Carbon Cube Kit package being sent containing eight ecojoiners and one wood surface will be:
- $((0.4 \times 8 \text{ ecojoiners}) + 1.4\text{kg}) = 4.6 \text{ kg}$
- To estimate the impact per package sent, we have done the calculations for a shipment from our factory in Situ Bondo, East Java to an address in Jakarta:

3 <http://www.newlifepaints.com/carbon-impact-of-waste-paint-the-stats>

4 According to [websitecarbon](#) our pages are lighter than 78-95% of pages on the internet. See: <https://www.websitecarbon.com/website/ecobricks-org-en-the-carbon-cube-chair-php/>

5 <https://twosidesna.org/US/The-Carbon-Footprint-of-Email-is-quite-large/>



- **Distance (road): ~922 km** driving distance Situbondo ↔ Jakarta
- **Road freight factor (rigid HGV 7.5–17 t, avg/50% laden): 0.2621 kg CO₂e/t-km** (UK Gov 2024 factors, widely used for logistics accounting and consistent with ISO 14083).
- **Optional first/last-mile** by motorcycle (pickup+drop total ~3 km) allocated by weight share:
 - Gasoline CO₂ factor **2.39 kg CO₂/L** (EMISI Indonesia). [WRI Indonesia+1](#)
 - Typical Indo scooter fuel economy ~**43.85 km/L** (Indonesian study). [SpringerOpen](#)
 - Parcel weight share assumption for courier load = **4.6 kg / (50 kg payload × 70% load-factor) ≈ 13.1%** (transparent assumption; adjust if you have better ops data).
- **Calculate Amount:**⁶
 - ~**1.13 kg CO₂e**
- Shipping total for 1000 units: **1130kg of CO₂e.**

6 Emissions were calculated following ISO 14083-consistent practice using activity data in tonne-kilometres and authoritative emission factors. Driving distance Situbondo–Jakarta was approximated at ~922 km (Rome2Rio). Road-freight emission intensity used the UK Government 2024 greenhouse-gas conversion factor for rigid HGV 7.5–17 t (average/50% laden): 0.2621 kg CO₂e per tonne-kilometre. Parcel mass 4.6 kg ⇒ 4.241 t-km, yielding 1.112 kg CO₂e for the trunk road leg. First/last-mile courier motorcycle was optionally included for a 3 km urban total, using Indonesia EMISI gasoline factor 2.39 kg CO₂/L and a representative scooter economy 43.85 km/L; emissions were allocated by parcel weight assuming 50 kg payload at 70% load factor, adding ~0.022 kg CO₂e. Total ≈ 1.13 kg CO₂e.



Product Process that Sequester CO₂

Bamboo Plantation

- The growth, cultivation and replanting of the bamboo used for producing our the carbon cube chair has an estimated effect of sequestering 0.51Kg of CO₂ per Kg of bamboo according to [a study of Asian bamboo products](#)⁷.
- As each 4FC ecojoiner (specifically for 600ml Aqua Bottles) has an average weight of 0.300Kg of bamboo, but is cut from a pole in which there are 700g of leftovers, we can calculate that each carbon cube chair, which requires 8 ecojoiners and a results in a total of bamboo use of :
- $(0.3 + 0.7) \times 8 \times 0.51 = 4.08 \text{ kg}$
- $\times 0.51 \times 1000 = \text{Plantation Sequestration per 1000 units: } \mathbf{4080 \text{ Kg}}$

Ecobrick Plastic Sequestration CO₂ Credit

6FC the carbon cube chair puts 12 600ml ecobricks to use. Ecobricks contain sequestered plastic. The average amount of plastic contained in a 600ml ecobrick is 234g of plastic⁸. The Plastic Sequestration inspired by the product (see Plastic section below) prevents this plastic from degrading and releasing its carbon into the atmosphere as CO₂. The GEA estimates that for each Kg of plastic 6.1 Kg of CO₂ are also sequestered.⁹

- Total CO₂ Credit from Plastic Sequestration for 1000 units:
- $1000 \times 12 \times 0.234 \times 6.1 = \mathbf{17,128.8 \text{ kg}}$

Replacing a similar Industrial Product

Products that replace similar plastic and/or industrially produced products can also claim a CO₂ credit.

7 <https://worldbamboo.net/wbcx/Keynotes/KeynotevanderLugt.pdf>

8 See Global Ecobrick Alliance Plastic Coefficients For ecological accounting
<https://ecobricks.org/en/coefficients.php>

9 [Www.ecobricks.org/why](http://www.ecobricks.org/why)



- the carbon cube chair provide a substitute for plastic furniture that is industrially produced. The initial use of the carbon cube chair is to make tables and chairs. Eight the carbon cube chair make one chair that can replace a standard polypropylene (PP) monobloc chair. recent Environmental Product Declaration for a standard polypropylene monobloc chair reports a cradle-to-grave carbon footprint of approximately **17.6 kg CO₂e per chair**, covering raw materials, manufacturing, transport, use, and end-of-life.¹⁰
 - Thus we can claim **17.6kg CO₂e** offset estimate per chair.
 - Offset impact for 1000 units: **17,600 kg of CO₂e**

10 Nardi S.p.A. (2021). *Environmental Product Declaration: Trill Armchair (Polypropylene Monobloc Chair)*. EPDItaly. Retrieved from <https://environdec.com/library/epd13383>



CO₂ Impacts				
CO₂ Produced per 1000 units				
Process	Details	Kg of CO₂	Units	Total
Production	Electric cutting, routing	23.4Kg	Batch	+ 79.5Kg
Materials	Glue, paint	13.58Kg	Batch	+ 27.16 kg
Server	Earthen Store purchase	0.05kg per order	1000	+0.5kg
Shipping	Indonesian Delivery	1.13kg per JNE shipment	1000	+ 1130kg
Total CO₂ Produced				+1237.16
CO₂ Sequestered				
Process	Details	CO₂/Kg	Units	Total
Bamboo	8Kg of bamboo per product @ 0.51Kg CO ₂ per Kg bamboo	-0.408Kg CO ₂	1000	- 4080 kg
Ecobrick Sequestration credit	12 x 234g Ecobricks	1.43.Kg CO ₂	12,000	- 17,128.8 kg
Replacement of Plastic Product	Replaces one monobloc plastic chair.	- 17.6kg CO ₂	1000	- 17,600 kg
Total Sequestered				- 38,808.8 kg
Total CO₂ Impact				-37571.64 kg
Per unit sequestration				-37.57 CO₂e



Plastic Impact

The plastic that is generated the product's life cycle has its own environmental impact. The disposal and recycling of plastic impacts the environment. Recycling can only process plastic several times-- each time plastic is recycled its value decreases until eventually it is no longer worth recycling. Consequently, all plastic, including recycled plastic, eventually ends up loose in the environment where it degrades into micro-plastics and chemicals that impact local ecologies. To estimate the products environmental impact, we record the net weight of all the plastic produced and consumed in its life-cycle.

the carbon cube chair also directly sponsor in the removal of plastic from the biosphere by their application of ecobricks. This is measure in Kg of plastic avoided and/or sequestered.¹¹

Plastic Production

Manufacturing

We estimate that we need one 250ml (@250g) can of epoxy glue per 25 the carbon cube chairs. We also will use 100 grams of pen and 3 glue bottles per 1000 units. We use a paper/plastic sticker for the the carbon cube chair that weighs 1g.

- Total plastic impact: 13.6 kg plastic per 1000 the carbon cube chair

Marketing / Labeling

We've managed to completely eliminate plastic from our packaging and labeling.

Shipping

- The Indonesian post office uses plastic coating labels for shipping and often insists on putting plastic tape on packages. We've designed our packaging to minimize this need. Per package plastic is about 3g.
- For 1000 units shipped = 3Kg plastic

¹¹ See Plastic Sequestration: <https://ecobricks.org/en/sequest.php>



Plastic Offsetting

Ecobrick Plastic Sequestration

6FC the carbon cube chair puts 12 600ml ecobricks to use. Ecobricks contain sequestered plastic. The average amount of plastic contained in a 600ml ecobrick is 234g of plastic¹².

- Consequently each chair results in 12 x 0.234 kg of plastic sequestered
- = **2.8kg total per chair**
- = **2800kg per 1000 units**

Replacing Plastic Equivalent Products

The carbon cube chair provide a substitutes for plastic furniture that is industrially produced. The initial use of the carbon cube chair is to make tables and chairs. One cube chair can replace a standard polypropylene (PP) monobloc chair that weighs on average ~2.5–5 kg of plastic depending on model/reinforcement. We will use the 4.7Kg value of the PP Trill Armchair¹³.

- Consequently, one carbon cube chair can claim an offset of 4.7kg of plastic production prevention
- **4700kg per 1000 units**

12 See Global Ecobrick Alliance Plastic Coefficients For ecological accounting
<https://ecobricks.org/en/coefficients.php>

13 See page 4: Nardi S.p.A. (2021). *Environmental Product Declaration: Trill Armchair (Polypropylene Monobloc Chair)*. EPDItaly. Retrieved from <https://environdec.com/library/epd13383>



Plastic Impacts				
Plastic Produced				
Process	Details	Kg Plastic/unit	Units	Total
Manufacturing	Glue bottles, epoxy, pen, sticker	13.6kg	batch	+13.6 kg
Marketing	none	0.000	-	-
Shipping	Tape, stickers	0.003		3Kg
Total Produced				+16.6 kg
Plastic Sequestered				
Process	Details	Kg Plastic/unit	Units	Total
Sponsored Ecobricking	12 ecobricks per Cube Chair	0.234kg	1000	2800 kg
Plastic Replacement	Plastic cups and tumblers not bought	4.7 Kg	1000	4700 Kg
Total Sequestered				-7500 kg
Total Plastic Impact				-7483.4 kg
Impact per unit				-7.5 kg of plastic